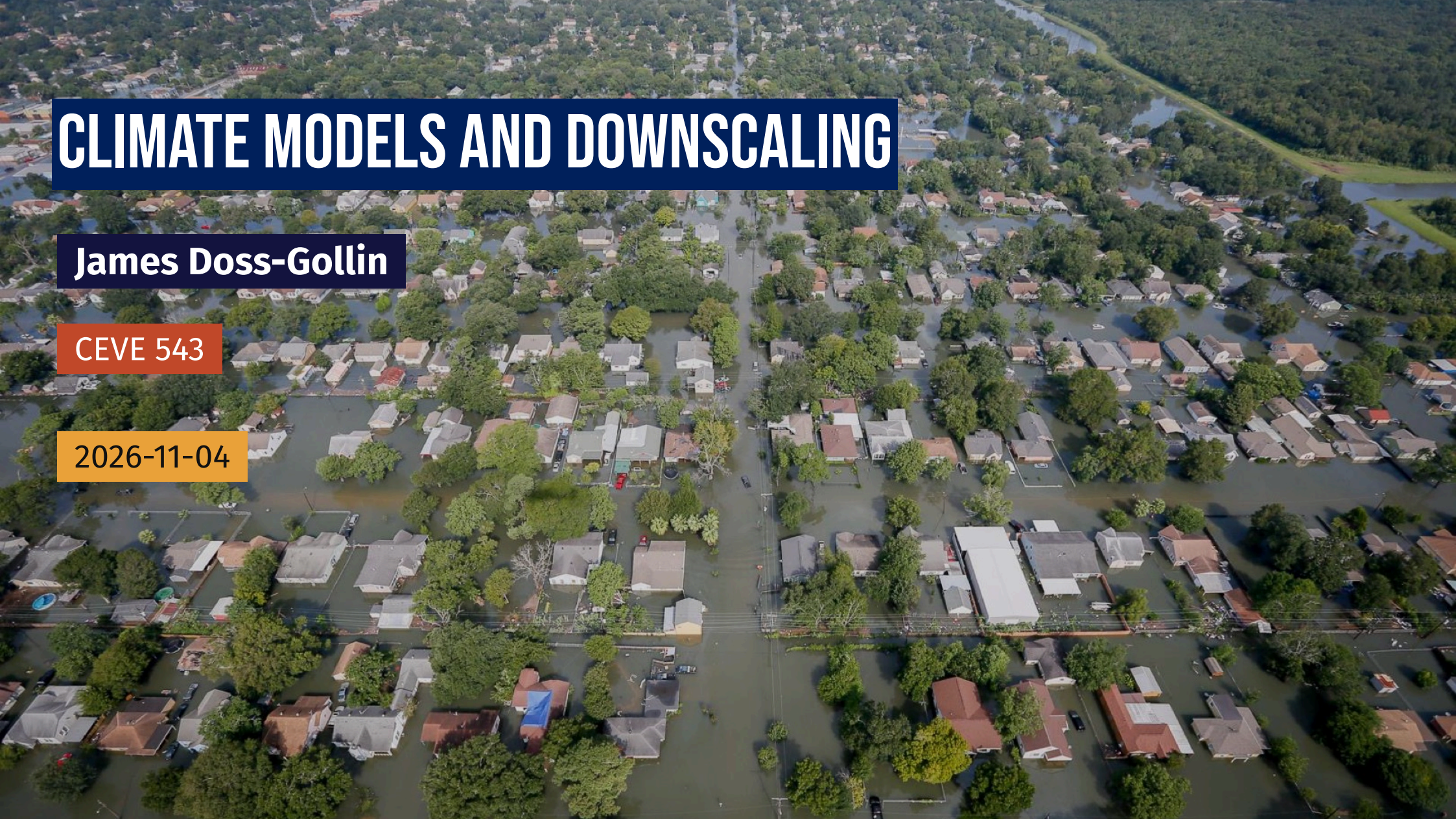
CLIMATE MODELS AND DOWNSCALING

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CEVE 543

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CAN A CLIMATE MODEL TELL AUSTIN HOW HARD IT WILL RAIN IN 2070?



Downtown Austin after the Memorial Day flood, May 2015 (City of Austin, public domain).

DRIZZLE BIAS

Coarser grids rain more often and more lightly, even in observations

Note

Figure shown in class: Tropical rain frequency and intensity against grid size, satellite and 24 models. Chen et al. (2021), Fig. 4.

LOW RESOLUTION OBSCURES IMPORTANT VARIABILITY

Statistical downscaling: learn a relationship between low- and high-resolution data.

Note

Figure shown in class: One day of observed rainfall at 1° and at 1/8°. Vandal et al. (2017), Fig. 1.

DYNAMICAL DOWNSCALING: REGIONAL MODELS INSIDE A GLOBAL ONE

Note

Figure shown in class: UK winter rainfall from a 300 km model, two regional models, and 5 km observations. Maraun et al. (2010), Fig. 1.

BIAS CORRECTION: A CLIMATE MODEL AGAINST ONE STATION IN KENYA

Note

Figure shown in class: ECHAM against a Katumani gauge: too many wet days, each too light. Ines & Hansen (2006), Fig. 1.

CORRECTING WEATHER FORECASTS AND CLIMATE PROJECTIONS

	Weather forecast	**Climate projection**
...		
Goal	$\hat{x}_t \approx x_t$: the corrected value matches what happened on day t	$p_{\hat{X}} \approx p_X$: the corrected values have the observed distribution
...		
Source of bias	right starting state, slightly wrong dynamics	a different dynamical system, running on its own

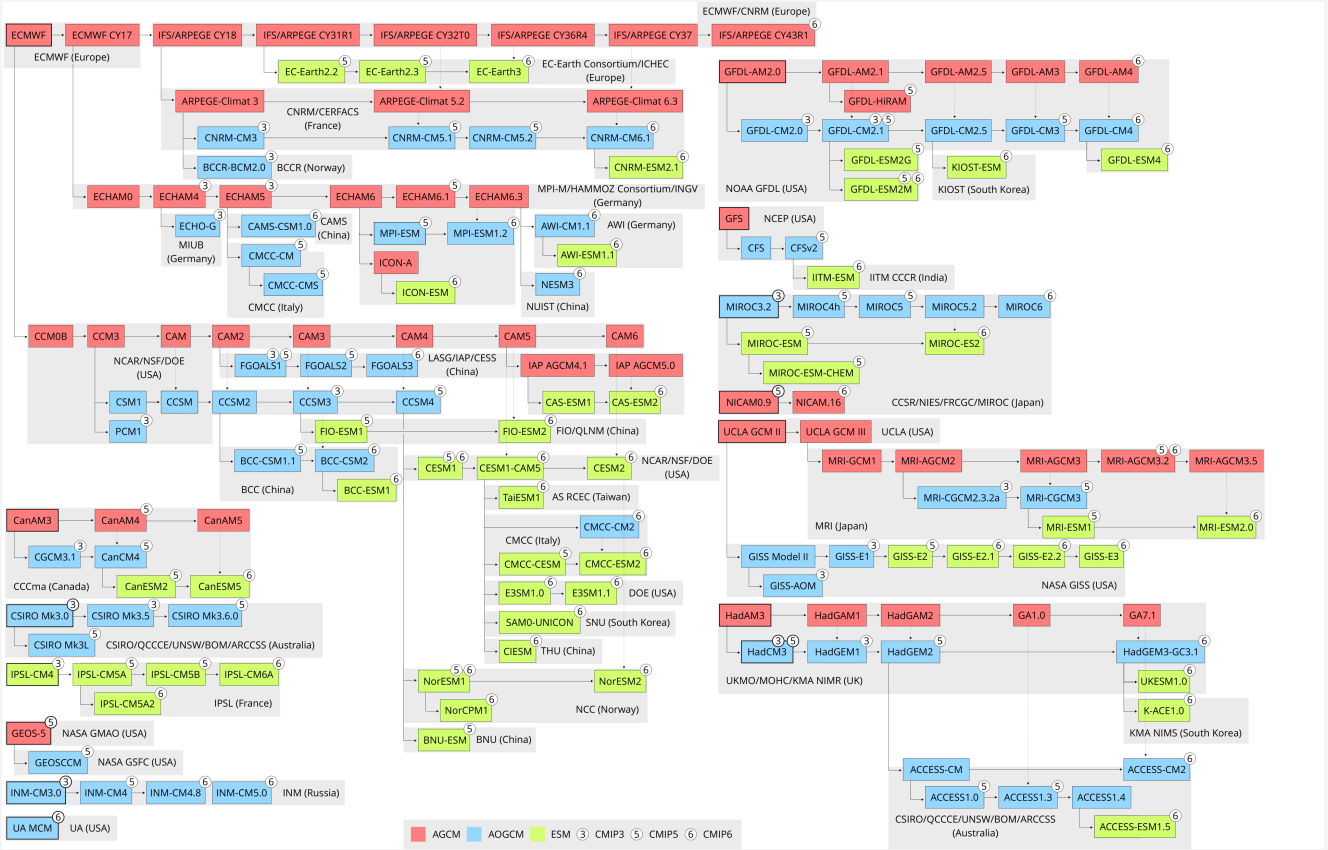
Reading CMIP data

READING A CMIP6 FILE PATH

```
CMIP6/ScenarioMIP/DKRZ/MPI-ESM1-2-HR/ssp585/r1i1p1f1/3hr/tas/gn/v20190710
```

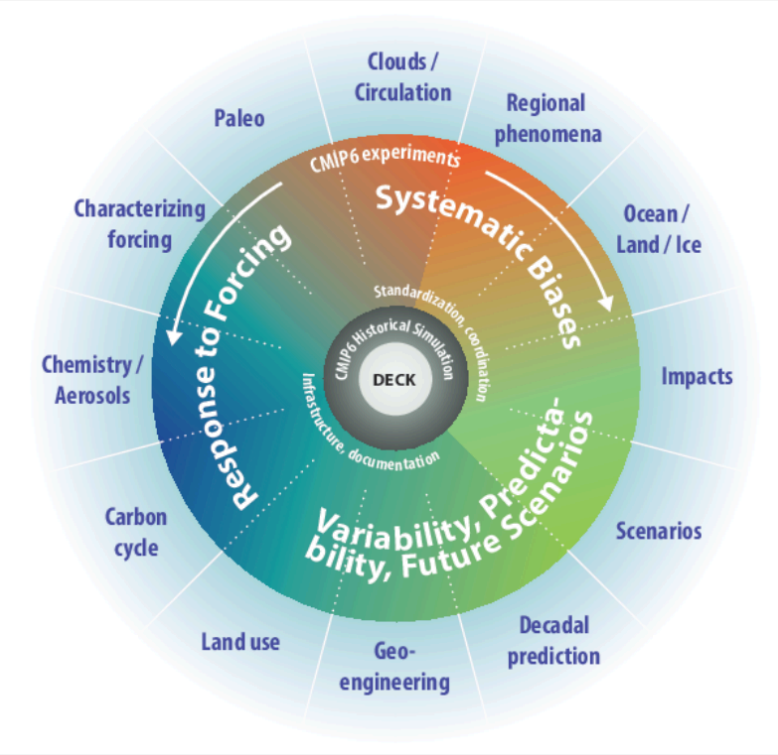
Part	Meaning
`DKRZ`	modeling center: German Climate Computing Center
`MPI-ESM1-2-HR`	model
`ssp585`	experiment: scenario SSP5-8.5
`r1i1p1f1`	ensemble member 1
`3hr`, `tas`	every 3 hours, near-surface air temperature

MODELING CENTERS AND MODELS



Code genealogy of CMIP3, CMIP5, and CMIP6 models, grouped by modeling center. Kuma et al. (2023) (CC BY 4.0).

EVERY MODEL RUNS THE SAME CORE EXPERIMENTS



The CMIP6 design: core experiments in the center, projects around them. Eyring et al. (2016), Fig. 2 (CC BY 3.0).

SCENARIOS: THE SAME FUTURES FOR EVERY MODEL

i Note

Figure shown in class: Global warming under five scenarios, CMIP6 models.
Intergovernmental Panel On Climate Change (2023), Fig. SPM.8a.

MODELS AND ENSEMBLES

- **Multi-model ensemble** (CMIP): many models, a few runs each; spread shows how models differ.
- **Initial-condition large ensemble**: one model, many runs with random noise (10^{-14} K) to study internal variability (Kay et al., 2015).
- **Perturbed-physics ensemble**: one model, uncertain parameters varied; spread shows parameter uncertainty (Stainforth et al., 2007).

Note

Figure shown in class: Winter temperature trends, 1979-2012: 30 runs of one model, their mean, observations. Kay et al. (2015), Fig. 4.